

SQL Fundamentals

Exercise 3

Q1

```
SELECT Product_name,  
       Price  
  
CASE  
  WHEN Price > 1000 THEN 'Expensive'  
  WHEN Price BETWEEN 100 AND 1000 THEN 'Mid-range'  
  WHEN Price < 100 THEN 'Budget'  
  
END AS Price_category  
FROM Products;
```

Outcome

Product_name	Price	Price_category
Laptop	1200.00	Expensive
Phone	850.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Midrange
Mouse	25.00	Budget

Q2.

```
SELECT Customer-name,  
        Amount
```

```
CASE
```

```
  WHEN Amount  $\geq$  1000 THEN 'High value'
```

```
  WHEN Amount BETWEEN 500 AND 999.99 THEN 'medium value'
```

```
  WHEN Amount  $<$  500 THEN 'low value'
```

```
END AS Order-value-category
```

```
FROM orders;
```

Outcome

Customer-name	Amount	Order-value-category
Alice	150.00	low value
Bob	860.00	Medium value
Charlie	999.99	Medium Value
Diana	45.50	Low value
Ethen	1200.00	High value

Q3

```
SELECT Emp_name,  
        department,  
        Salary
```

```
CASE
```

```
    WHEN department = 'IT' AND Salary > 80000 THEN 'Senior IT'
```

```
    WHEN department = 'HR' AND Salary > 55000 THEN 'Experienced HR'
```

```
    ELSE 'Staff'
```

```
END AS Position_level
```

```
FROM Employees;
```

Outcome

emp_name	department	Salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	85000	Staff

Q4

```
SELECT Student_name,  
       Score,
```

```
CASE
```

```
  WHEN Score >= 90 THEN 'A'
```

```
  WHEN Score BETWEEN 80 AND 89 THEN 'B'
```

```
  WHEN Score BETWEEN 70 AND 79 THEN 'C'
```

```
  WHEN Score BETWEEN 60 AND 69 THEN 'D'
```

```
  WHEN Score < 60 THEN 'F'
```

```
END AS grade
```

```
FROM Students;
```

Outcome

Student_name	Score	grade
Anng	92	A
Ben	76	C
Cara	59	F
David	83	B
Eller	68	D

Q5

SELECT Delivery - ID,
delivery - time - minutes

CASE

WHEN delivery - time - minutes ≤ 30 THEN 'Fast'

WHEN delivery - time - minutes BETWEEN 31 AND 60 THEN 'on time'

WHEN delivery - time - minutes > 60 THEN 'Late'

END AS performance

FROM deliveries;

Outcome

delivery - id	delivery - time - minutes	performance
1	25	on time
2	80	Late
3	30	fast
4	65	Late
5	60	Late

Q6

```
SELECT Issue_type,  
       Priority  
CASE Priority  
  WHEN 3 THEN 'High'  
  WHEN 2 THEN 'Medium'  
  WHEN 1 THEN 'Low'  
END AS Priority_label  
FROM tickets;
```

Outcome

Issue_type	Priority	Priority_label
Login\$ Issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

Q7

```

SELECT Student - id
       days - present
CASE Attendance - percentage
  WHEN days - present * 100.0 / total - days >= 90 THEN 'Excellent'
  WHEN days - present * 100.0 / total - days BETWEEN 75 AND 89.99
    THEN 'Good'
  WHEN days - present * 100.0 / total - days < 75 THEN 'Needs
    improving'
END AS Attendance - Status
FROM attendance.
  
```

Outcome

Student - id	Attendance - percentage	attendance - Status
1	90.00	Excellent
2	60.00	Needs improving
3	96.00	Excellent
4	50.00	Needs improving
5	100.00	Excellent.

Q8.

```
SELECT product_id,  
       stock_qty
```

```
CASE
```

```
  WHEN stock_qty = 0 THEN 'Out of Stock'
```

```
  WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
```

```
  WHEN stock_qty > 5 THEN 'In Stock'
```

```
END AS stock_status
```

```
FROM Product_Inventory;
```

Outcome

Product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock
6		

Q9.

```
SELECT Subject,  
        Enrolled-Students,
```

```
CASE
```

```
  WHEN Enrolled-Students >= THEN 'Large
```

```
  WHEN Enrolled-Students BETWEEN 10 AND 24 THEN 'medium'
```

```
  WHEN Enrolled-Students < 10 THEN 'Small'
```

```
END AS Class-Size-Category
```

```
FROM classes;
```

Outcome

Subject	Enrolled-Student	Class-Size-Category
Maths	30	Large
English	25	Large

Q10

```
SELECT Payment - Id,  
       Payment - method,  
       amount
```

CASE

WHEN Payment - method = 'cash' AND amount >= 200

ELSE 'Not - Eligible'

END AS discount - eligibility

FROM Payments;

Outcome

Payment - Id	Payment - method	amount	discount - Eligability
1	Card	50.00	Not eligible
2	Cash	200.00	Eligable for discount
3	Card	150.00	Not Eligable
4	PayPal	75.00	Not eligible
5	Cash	300.00	Eligable for discount